

Algorithmic Pseudocode Sample 8

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

- 1: **Input:** n independent realizations of the p -random vector $\mathbf{X}, \mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(n)}$; a topological ordering Ord , (and a stopping level m).
 - 2: **Output:** An estimated DAG $\hat{G}$.
 - 3: Form the complete undirected graph $\tilde{G}$ on the vertex set V .
 - 4: Orient edges on $\tilde{G}$ respecting the topological ordering to form DAG G' .
 - 5: $l = -1; \quad \hat{G} = G'$
 - 6: **repeat**
 - 7: $l = l + 1$
 - 8: **for** all vertices $s \in V$, **do**
 - 9: let $\mathbf{K}_s = pa(s)$
 - 10: **end for**
 - 11: **repeat**
 - 12: Select a (new) edge $t \rightarrow s$ in $\hat{G}$ such that
 - 13: $|\mathbf{K}_s \setminus \{t\}| \geq l$.
 - 14: **repeat**
 - 15: choose a (new) set $\mathbf{S} \subset \mathbf{K}_s \setminus \{t\}$ with $|\mathbf{S}| = l$.
 - 16: **if** $H_0 : \theta_{st|\mathbf{K}} = 0$ not rejected
 - 17: delete edge $t \rightarrow s$ from $\hat{G}$
 - 18: **end if**
 - 19: **until** edge $t \rightarrow s$ is deleted or all $\mathbf{S} \subset \mathbf{K}_s \setminus \{t\}$ with $|\mathbf{S}| = l$ have been considered.
 - 20: **until** all edge $t \rightarrow s$ in $\hat{G}$ such that $|\mathbf{K}_s \setminus \{t\}| \geq l$ and $\mathbf{S} \subset \mathbf{K}_s \setminus \{t\}$ with $|\mathbf{S}| = l$ have been tested for conditional independence.
 - 21: **until** $l = m$ or for each edge $t \rightarrow s$ in $\hat{G}$: $|\mathbf{K}_s \setminus \{t\}| < l$.
-